



**ANNA UNIVERSITY REGIONAL CAMPUS
COIMBATORE**

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

PROJECT TITLE

**SIMULATION OF PATCH ANTENNA WITH METAMATERIAL
SUPERSTRATE**

EC3452 -Electromagnetic fields

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INTRODUCTION:

Patch antennas are widely used in wireless communication systems due to their low profile, lightweight, and ease of fabrication. However, they often suffer from narrow bandwidth and low gain. Metamaterials, which are artificially structured materials with unique electromagnetic properties, can be used to improve the performance of patch antennas. This project focuses on the design and simulation of a patch antenna with a metamaterial superstrate using openEMS and GNU Octave.

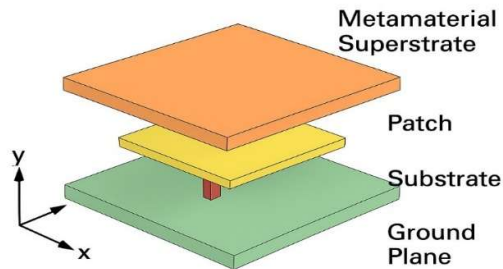
OBJECTIVE:

- Design a rectangular Patch antenna operating at 2.45GHz.
- To enhance its performance using a metamaterial superstrate.
- To analyse the reflection coefficient(S_{11}), impedance, and radiation pattern.

TOOLS USED:

- GNU Octave: Used for writing and running simulation scripts.
- OpenEMS : A free and open-source electromagnetic field solver using the Finite Difference Time Domain (FDTD)
- CSXCAD: Geometry and mesh definition.
- ParaView: 3D field visualization.

MICROSTRIP PATCH ANTENNA WITH METAMATERIAL



GROUND PLANE:

- Bottom most layer.
- Perfect Electric Conductor.
- Metal(Copper like surface).
- Acts as RF ground, reflects radiation.

SUBSTRATE:

- Above ground.
- Dielectric(Eg Rogers).
- Epsilon =12 or 3.38.
- Supports the patch,affects bandwidth & impedance.

PATCH ANTENNA:

- Above substrate.
- Perfect Electric Material.(Copper patch)
- Resonant structure (typically rectangular)
- Radiates EM waves,main active element.

METAMATERIAL SUPERSTRATE:

- Top most layer.
- Artificial Material(Metamaterial)
- Epsilon = -2, Mu = -1(negative)
- Enhance gain, directivity and beam shaping.

FEED:

- ❖ Type : Lumped port
- ❖ Position : -5.5mm from center
- ❖ Width : 2mm
- ❖ Resistance : 50 Ohms

PURPOSE:

- To improve gain and directivity.
- Acts like a lens focusing radiation from the patch.
- Can introduce negative permittivity or permeability at certain frequencies

SIMULATION SETUP:

1. FREQUENCY RANGE:

- Center frequency: 2.45 GHz
- Bandwidth: 1 GHz

2. BOUNDARY CONDITIONS:

Perfectly Matched Layer (PML) is used on all six sides to absorb outgoing waves and mimic open space.

PROGRAM:

Close all

clear

clc

%% switches & options

postprocessing_only = 0;

draw_3d_pattern = 1;

use_pml = 1;

openEMS_opts = '';

%% physical constants and units

physical_constants;

unit = 1e-3; % mm

%% antenna parameters

patch.width = 32.86;

patch.length = 41.37;

substrate.epsR = 3.38;

substrate.kappa = $1e-3 * 2 * \pi * 2.45e9 * \epsilon_0$ * substrate.epsR;

substrate.width = 60;

substrate.length = 60;

substrate.thickness = 1.524;

substrate.cells = 4;

meta.epsR = 12;

meta.thickness = 3;

meta.height_above_patch = 5;

feed.pos = -5.5;

```

feed.width = 2;

feed.R = 50;

SimBox = [100 100 50]; % larger z for superstrate

%% simulation folder

Sim_Path = 'tmp_metamaterial_patch';

Sim_CSX = 'patch_meta.xml';

if (postprocessing_only==0)

    rmdir(Sim_Path, 's');

    mkdir(Sim_Path);

end

%% FDTD

f0 = 2.45e9;

fc = 1e9;

max_timesteps = 30000;

min_decrement = 1e-5;

FDTD = InitFDTD('NrTS', max_timesteps, 'EndCriteria', min_decrement);

FDTD = SetGaussExcite(FDTD, f0, fc);

BC = repmat({'MUR'}, 1, 6);

if use_pml

    BC = repmat({'PML_8'}, 1, 6);

End

FDTD = SetBoundaryCond(FDTD, BC);

%% mesh

max_res = c0 / (f0 + fc) / unit / 20;

CSX = InitCSX();

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mesh.x = SmoothMeshLines([-SimBox(1)/2 -patch.width/2 feed.pos
patch.width/2 SimBox(1)/2], max_res, 1.4);

mesh.y = SmoothMeshLines([-SimBox(2)/2 -patch.length/2 patch.length/2
SimBox(2)/2], max_res, 1.4);

mesh.z = [ ...
    linspace(-SimBox(3)/2, 0, 5), ...
    linspace(0, substrate.thickness, substrate.cells), ...
    linspace(substrate.thickness + meta.height_above_patch, ...
        substrate.thickness + meta.height_above_patch + meta.thickness, 4), ...
    SimBox(3)];

mesh.z = SmoothMeshLines(mesh.z, max_res, 1.4);

mesh = AddPML(mesh, [8 8 8 8 8]);

CSX = DefineRectGrid(CSX, unit, mesh);

%% Patch

CSX = AddMetal(CSX, 'Patch');

CSX = AddBox(CSX, 'Patch', 10, ...
    [-patch.width/2, -patch.length/2, substrate.thickness], ...
    [ patch.width/2,  patch.length/2, substrate.thickness]);

%% Substrate

CSX = AddMaterial(CSX, 'Substrate');

CSX = SetMaterialProperty(CSX, 'Substrate', 'Epsilon', substrate.epsR, 'Kappa',
substrate.kappa);

CSX = AddBox(CSX, 'Substrate', 0, ...
    [-substrate.width/2, -substrate.length/2, 0], ...
    [ substrate.width/2,  substrate.length/2, substrate.thickness]);

%% Ground plane

```

```

CSX = AddMetal(CSX, 'Ground');
CSX = AddBox(CSX, 'Ground', 10, ...
    [-substrate.width/2, -substrate.length/2, 0], ...
    [ substrate.width/2, substrate.length/2, 0]);
%% Metamaterial superstrate
meta_z0 = substrate.thickness + meta.height_above_patch;
CSX = AddMaterial(CSX, 'Metamaterial');
CSX = SetMaterialProperty(CSX, 'Metamaterial', 'Epsilon', meta.epsR);
CSX = AddBox(CSX, 'Metamaterial', 0, ...
    [-patch.width/2, -patch.length/2, meta_z0], ...
    [ patch.width/2, patch.length/2, meta_z0 + meta.thickness]);
%% Lumped port
start = [feed.pos -feed.width/2 0];
stop  = [feed.pos +feed.width/2 substrate.thickness];
[CSX, port] = AddLumpedPort(CSX, 5, 1, feed.R, start, stop, [0 0 1], true);
%% Magnetic field dump
CSX = AddDump(CSX, 'Ht_', 'DumpType', 1, 'DumpMode', 2);
CSX = AddBox(CSX, 'Ht_', 0, ...
    [-patch.width, -patch.length, substrate.thickness + 1], ...
    [ patch.width, patch.length, substrate.thickness + 1]);
%% NF2FF box
[CSX, nf2ff] = CreateNF2FFBox(CSX, 'nf2ff', -SimBox/2, SimBox/2);
%% Run simulation
if postprocessing_only == 0
    WriteOpenEMS([Sim_Path '/' Sim_CSX], FDTD, CSX);

```



```

CSXGeomPlot([Sim_Path '/' Sim_CSX]);

RunOpenEMS(Sim_Path, Sim_CSX, openEMS_opts);

end

%% Postprocessing

freq = linspace(f0 - fc, f0 + fc, 501);

U = ReadUI({'port_ut1','et'}, Sim_Path, freq);

I = ReadUI('port_it1', Sim_Path, freq);

%% Time domain voltage

figure;

[ax, h1, h2] = plotyy(U.TD{1}.t/1e-9, U.TD{1}.val, U.TD{2}.t/1e-9, U.TD{2}.val);

set(h1, 'Linewidth', 2, 'Color', [1 0 0]);

set(h2, 'Linewidth', 2, 'Color', [0 0 0]);

grid on;

title('Time domain voltage');

xlabel('Time t / ns');

ylabel(ax(1), 'Voltage ut1 / V');

ylabel(ax(2), 'Voltage et / V');

%% Feed impedance

figure;

Zin = U.FD{1}.val ./ I.FD{1}.val;

plot(freq/1e6, real(Zin), 'k-', 'LineWidth', 2); hold on;

plot(freq/1e6, imag(Zin), 'r--', 'LineWidth', 2);

grid on;

title('Feed point impedance');

xlabel('Frequency f / MHz'); ylabel('Impedance Z_{in} / Ohm');

```

```

legend('Real', 'Imag');

%% Reflection coefficient S11

figure;

uf_inc = 0.5 * (U.FD{1}.val + I.FD{1}.val * feed.R);

uf_ref = U.FD{1}.val - uf_inc;

s11 = uf_ref ./ uf_inc;

plot(freq/1e6, 20*log10(abs(s11)), 'k-', 'LineWidth', 2);

grid on;

title('Reflection coefficient S_{11}');

xlabel('Frequency f / MHz'); ylabel('|S_{11}| / dB');

P_in = 0.5 * U.FD{1}.val .* conj(I.FD{1}.val);

%% NF2FF plots

f_res_ind = find(abs(s11) == min(abs(s11)));

f_res = freq(f_res_ind);

thetaRange = (0:2:359) - 180;

phiRange = [0 90];

nf2ff = CalcNF2FF(nf2ff, Sim_Path, f_res, thetaRange*pi/180,
phiRange*pi/180);

Dlog = 10*log10(nf2ff.Dmax);

disp(['Radiated power: Prad = ', num2str(nf2ff.Prad), ' W']);

disp(['Directivity: Dmax = ', num2str(Dlog), ' dBi']);

disp(['Efficiency: nu_rad = ', num2str(100*nf2ff.Prad ./ real(P_in(f_res_ind))), '
%']);

figure;

plotFFdB(nf2ff, 'xaxis', 'theta', 'param', [1 2]);

drawnow;

```

```

if draw_3d_pattern

    phiRange = 0:2:360;

    thetaRange = 0:2:180;

    nf2ff = CalcNF2FF(nf2ff, Sim_Path, f_res, thetaRange*pi/180,
    phiRange*pi/180, 'Verbose', 2, 'Outfile', 'nf2ff_3D.h5');

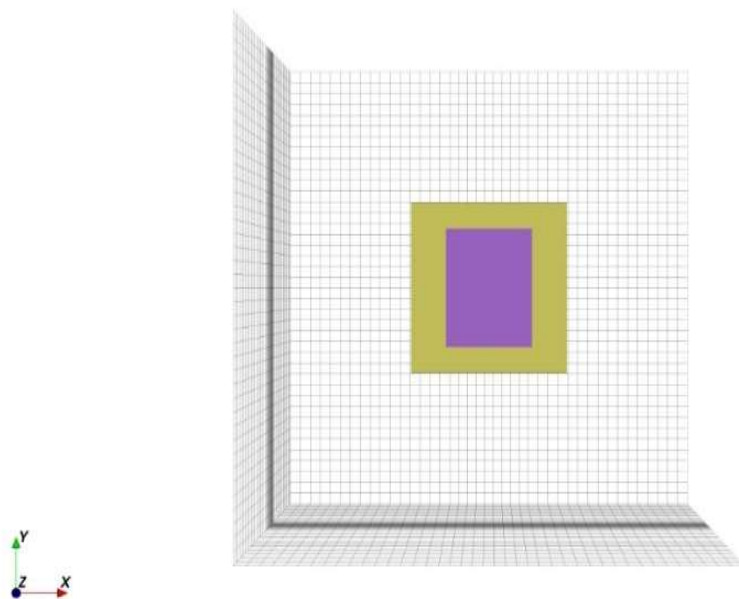
    figure;

    plotFF3D(nf2ff);

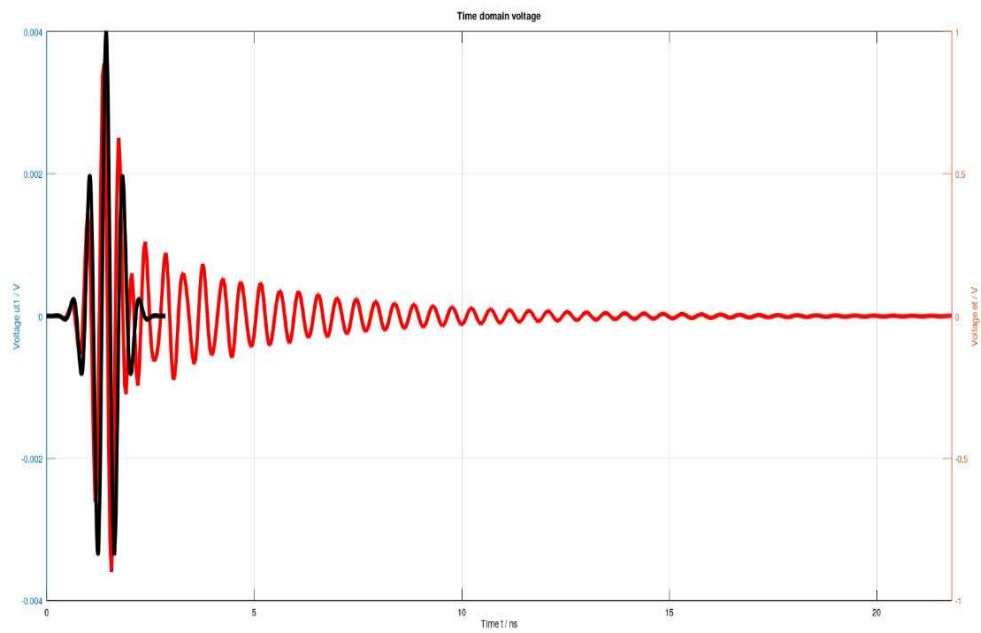
end

```

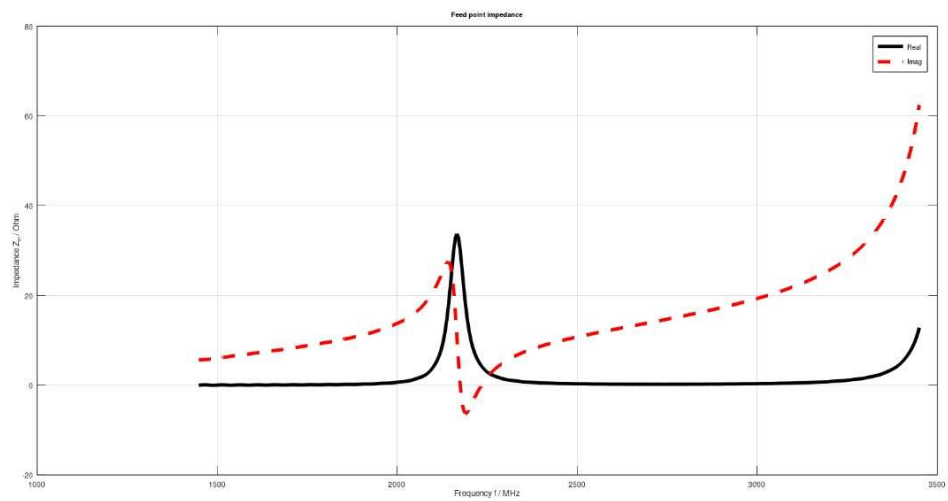
OUTPUT:



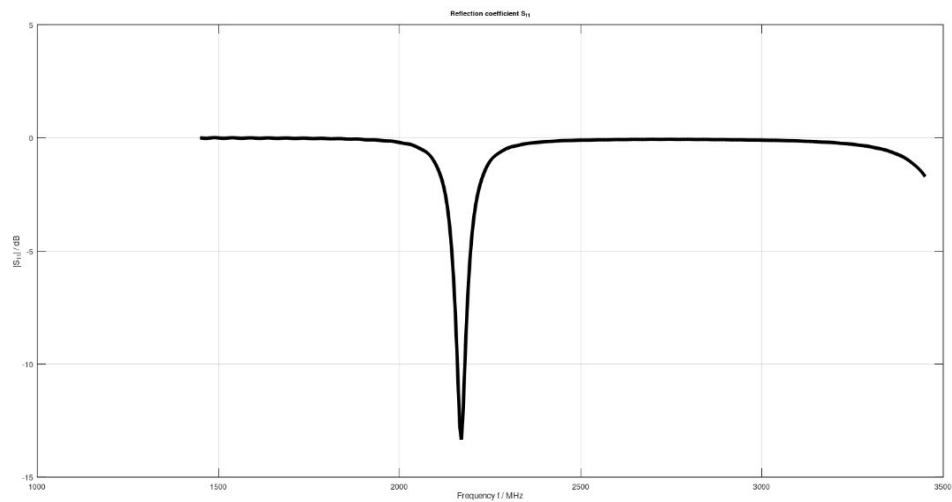
AFTER SIMULATION:
TIME VOLTAGE GRAPH



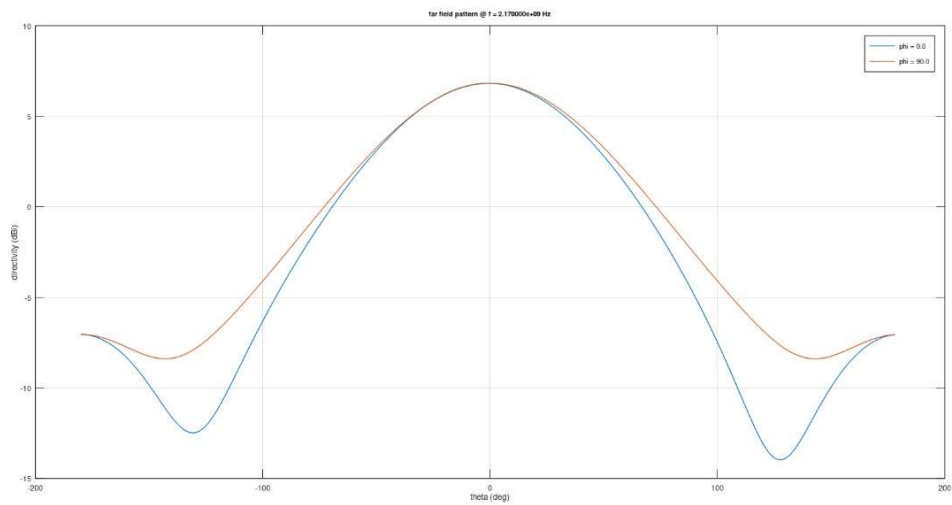
IMPEDANCE GRAPH:



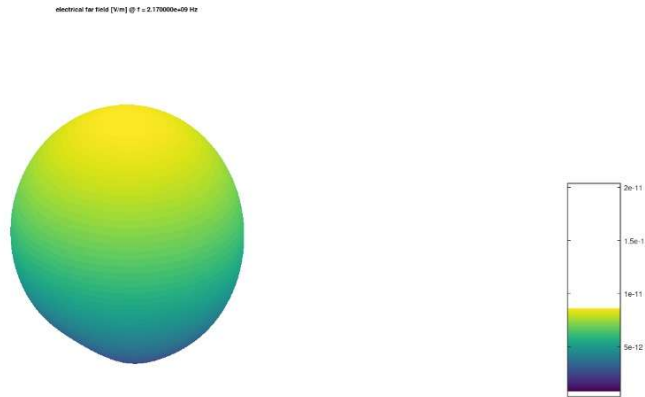
REFLECTION COEFFICIENT GRAPH:



FAR FIELD GRAPH:



3D VISUALIZATION OF ANTENNA:



SIMULATION RESULTS:

1) REFLECTION COEFFICIENT:

- Return loss(S11) at 2.45GHz: below -10 db indicates good matching.
- Bandwidth: approximately 100-200MHz depending on tuning.

2) IMPEDENCE:

Close to 50 ohms at resonance, indicating proper matching.

3) RADIATION PATTERN:

- Main lobe directed in +z direction.
- Metamaterial superstrate enhances the directivity.
- Gain is expected to improve due to focused radiation.

CONCLUSION:

The simulation results show that the inclusion of a metamaterial superstrate with a dielectric constant of 12 enhances the antenna's performance by improving impedance matching and radiation

characteristics. This confirms the potential of metamaterials in advanced antenna design for wireless applications.